

Research Notes on Uncertainty quantification

Uncertainty quantification

>> Section 1

Simulation-based methods: Monte Carlo simulations, importance sampling, adaptive sampling, etc. General surrogate-based methods: In a non-intrusive approach, a surrogate model is learnt in order to replace the experiment or the simulation with a cheap and fast approximation. Surrogate-based methods can also be employed in a fully Bayesian fashion. This approach has proven particularly powerful when the cost of sampling, e.g. computationally expensive simulations, is prohibitively high. Local expansion-based methods: Taylor series, perturbation method, etc. These methods have advantages when dealing with relatively small input variability and outputs that don't express high nonlinearity. These linear or linearized methods are detailed in the article [Uncertainty propagation](#). Functional expansion-based methods: Neumann expansion, orthogonal or Karhunen–Loeve expansions (KLE), with polynomial chaos expansion (PCE) and wavelet expansions as special cases. Most probable point (MPP)-based methods: first-order reliability method (FORM) and second-order reliability method (SORM). Numerical integration-based methods: Full factorial numerical integration (FFNI) and dimension reduction (DR). For non-probabilistic approaches, interval analysis, Fuzzy theory, Possibility theory and evidence theory are among the most widely used. The probabilistic approach is considered as the most rigorous approach to uncertainty analysis in engineering design due to its consistency with the theory of decision analysis. Its cornerstone is the calculation of probability density functions for sampling statistics. This can be performed rigorously for random variables that are obtainable as transformations of Gaussian variables, leading to exact confidence intervals.

>> Section 2

Modular approach An approach to inverse uncertainty quantification is the modular Bayesian approach. The modular Bayesian approach derives its name from its four-module procedure. Apart from the current available data, a prior distribution of unknown parameters should be assigned.

>> Section 3

The opinion triangles represent barycentric coordinate systems, where a subjective opinion is visualized as a blue dot, with a blue frame for associated values for Belief, Disbelief and E-uncertainty (epistemic), where $\text{Belief} + \text{Disbelief} + \text{E-uncertainty} = 1$. The blue frame also shows base rate (prior) and probability values. A-uncertainty (aleatoric) is greatest when the probability is $P = 0.5$. The Beta probability density function to the right of the triangles

visualizes where on the scale $[0, 1]$ the probability has the greatest density, i.e. where the probability presumably lies. See the subjective opinion online demo of uncertainty visualization.

>> Section 4

To evaluate low-order moments of the outputs, i.e. mean and variance. To evaluate the reliability of the outputs. This is especially useful in reliability engineering where outputs of a system are usually closely related to the performance of the system. To assess the complete probability distribution of the outputs. This is useful in the scenario of utility optimization where the complete distribution is used to calculate the utility. To estimate uncertainty in values, that cannot be directly measured, when calculated with experimentally measurable values. This is especially useful in experimental physics and chemistry contexts.

>> Section 5

$$R_m(\mathbf{x}, \boldsymbol{\theta}, (\mathbf{x}', \boldsymbol{\theta}')) = \exp \left\{ - \sum_{k=1}^d \omega_k m(\mathbf{x} - \mathbf{x}')^2 \right\} \exp \left\{ - \sum_{k=1}^r \omega_d + k m(\boldsymbol{\theta}_k - \boldsymbol{\theta}_k')^2 \right\}.$$
$$\{ \displaystyle R^m(\mathbf{x}, \boldsymbol{\theta}, (\mathbf{x}', \boldsymbol{\theta}')) = \exp \left\{ - \sum_{k=1}^d \omega_k m(\mathbf{x} - \mathbf{x}')^2 \right\} \exp \left\{ - \sum_{k=1}^r \omega_d + k m(\boldsymbol{\theta}_k - \boldsymbol{\theta}_k')^2 \right\}.$$

>> Section 6

$$y_m(\mathbf{x}, \boldsymbol{\theta}) \sim \mathcal{GP}(\mathbf{h}_m(\cdot), \boldsymbol{\Sigma}_m, \sigma_m^2, R_m(\cdot, \cdot))$$
$$\{ \displaystyle y^m(\mathbf{x}, \boldsymbol{\theta}) \sim \mathcal{GP}(\mathbf{h}^m(\cdot), \boldsymbol{\Sigma}^m, \sigma^m^2, R^m(\cdot, \cdot))$$

>> Section 7

Derive the posterior distribution $p(\boldsymbol{\theta}, \boldsymbol{\phi} \mid \text{data})$
$$\{ \displaystyle p(\boldsymbol{\theta}, \boldsymbol{\phi} \mid \text{data})$$

; Integrate $\boldsymbol{\phi}$
$$\{ \displaystyle p(\boldsymbol{\theta} \mid \text{data})$$

This single step accomplishes the calibration; Prediction of the experimental response and discrepancy function. However, the approach has significant drawbacks:

>> Section 8

Mathematical perspective In mathematics, uncertainty is often characterized in terms of a probability distribution. From that perspective, aleatoric uncertainty means that the probability distribution does not unambiguously point to a specific outcome or not being able to predict what a random sample drawn from a probability distribution will be, while epistemic uncertainty means that one does not

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know the relevant probability distribution because one lacks a basis for determining it. The figure below shows four different examples of aleatoric and epistemic uncertainty, visualized in two mathematically equivalent ways consisting of a triangle of subjective opinions from subjective logic and as a Beta probability density function.

>> Section 9

Bayesian Several methodologies for inverse uncertainty quantification exist under the Bayesian framework. The most complicated direction is to aim at solving problems with both bias correction and parameter calibration. The challenges of such problems include not only the influences from model inadequacy and parameter uncertainty, but also the lack of data from both computer simulations and experiments. A common situation is that the input settings are not the same over experiments and simulations. Another common situation is that parameters derived from experiments are input to simulations. For computationally expensive simulations, then often a surrogate model, e.g. a Gaussian process or a Polynomial Chaos Expansion, is necessary, defining an inverse problem for finding the surrogate model that best approximates the simulations.

>> Section 10

d is the dimension of input variables, and r is the dimension of unknown parameters. While $\mathbf{h}^m(\cdot)$ is pre-defined, $\{\beta_m, \sigma_m, \omega_k^m, k=1, \dots, d+r\}$, known as hyperparameters of the GP model, need to be estimated via maximum likelihood estimation (MLE). This module can be considered as a generalized kriging method.

>> Section 11

$$p(\boldsymbol{\theta} \mid \text{data}, \boldsymbol{\phi}) \propto p(\text{data} \mid \boldsymbol{\theta}, \boldsymbol{\phi}) p(\boldsymbol{\theta})$$